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COS30031 – Task 2 – 9th August 2023

Release Notes:

- C++ Recourse Links that I found and used were:
 - C++ Reference / Cheat Sheet -
<https://www.codewithharry.com/blogpost/cpp-cheatsheet/>
 - Visual Studio Code Keyboard Short-Cut Reference Sheet –
<https://code.visualstudio.com/shortcuts/keyboard-shortcuts-windows.pdf>
 - Online Code Debugging Tool –
<https://www.onlinegdb.com/>
- Lab Demo Code Snippets

```
// 1. created a function that takes two values and prints them.
void PrintTwoValues(int a, int b)
{
    cout << "Int 1 : " << a << " "
    << "Int 2 : " << b << endl;
}
```

Figure 1 Function that prints 2 values.

```
// 2. created a function that alters a value and returns them.
int AlterValue(int a)
{
    a *= 4;
    return a;
}
```

Figure 2 Function that alters a value.

```
// 3. declare a pointer to a variable and demonstrate the setting and reading of the value in the variable using the pointer
void demoPointer()
{
    int a = 3;
    int *ptr = &a;
    cout << "Original Value of a : " << a << endl;
    cout << "Value of a using pointer : " << *ptr << endl;
}
```

Figure 3 Function that demonstrates pointers and reading values in a pointer.

```
// 4. Create a function that uses a for loop to print all odd numbers between 0 and 20 on a single text line of text output, formatted with spaces. Call this from main.
void PrintOddNumbers()
{
    cout << "Odd Numbers : ";
    for (int i = 0; i < 20; i++)
    {
        if (i % 2 != 0)
        {
            cout << i << " ";
        }
    }
    cout << endl;
}
```

Figure 4 Function that prints odd numbers.

```
// 5. created a function that creates a 1 dimensional array of 5 integers, then fills the array with random int values, prints the contents of the array.
void PrintArray()
{
    You, 2 days ago • created a function that fills up an array ...
    cout << "Array : ";
    int arr[5];
    for (int i = 0; i < 5; i++)
    {
        arr[i] = rand() % 100;
        cout << arr[i] << " ";
    }
    cout << endl;
}
```

Figure 5 Function that prints a 1-dimensional array that contains 5 integers.

```
// 6. created a function that splits the line of text such as "this has spaces in it"
void PrintSplitTextbySpaces(string text)
{
    cout << "Split text: "; You, 2 days ago • created a function to split text th
    size_t startpos = 0;
    size_t endpos = text.find(" ");
    while (endpos != string::npos)
    {
        cout << text.substr(startpos, endpos - startpos) << endl;
        startpos = endpos + 1;
        endpos = text.find(" ", startpos);
    }
    cout << text.substr(startpos) << endl;
}
```

Figure 6 Function that splits text that have spaces between them into separate text lines.

```

// 7. declare a simple class that has least one private variable, public variable and public method. create an instance of the class and demo its use
You, 2 days ago | 1 author (You)
class demoClass
{
private:
    int privateVar;    You, 2 days ago + created a simple class

public:
    int publicVar;

    demoClass()
    {
        privateVar = 0;
        publicVar = 0;
    }

    void setPrivateVar(int value)
    {
        privateVar = value;
    }

    int getPrivateVar()
    {
        return privateVar;
    }

    void printValues()
    {
        cout << "Private Var : " << privateVar << endl;
        cout << "Public Var : " << publicVar << endl;
    }
};

```

Figure 7 Function that demonstrates how classes work.